

This chapter deals with the gross anatomy of the structures that lie within the vertebral canal and their extensions through the intervertebral foramina, the spinal nerve or radicular ('root') canals. The internal organization of the spinal cord is described in [Chapter 20](#).

The spinal cord and its blood vessels and nerve roots lie within a meningeal sheath, the theca, which occupies the central zone of the vertebral canal and extends from the foramen magnum, where it is in continuity with the meningeal coverings of the brain, to the level of the second sacral vertebra in the adult. Distal to this level, the dura extends as a fine cord, the filum terminale externum, which fuses with the posterior periosteum of the first coccygeal segment. Tubular prolongations of the dural sheath extend around the spinal roots and nerves into the lateral zones of the vertebral canal and out into the root canals, eventually fusing with the epineurium of the spinal nerves. Between the theca and the walls of the vertebral canal is the epidural (spinal extradural) space, which is loosely filled with fat, connective tissue containing small arteries and lymphatics, and an important venous plexus. Three-dimensional appreciation of the anatomy of the spinal theca and its surroundings is essential for the efficient management of spinal pain and of spinal injuries, tumours and infections. Equally significant clinically is the anatomy of the often-precarious blood supply of the spinal cord and its associated structures. The increasing application and refinement of diagnostic imaging and endoscopic procedures lend a new importance to topographical detail here.

SPINAL CORD

The spinal cord occupies the superior two-thirds of the vertebral canal ([Figs 45.1–45.2](#)). It is continuous cranially with the medulla oblongata, just below the level of the foramen magnum, at the upper border of the atlas. It terminates caudally as the conus medullaris. During development, the vertebral column elongates more rapidly than the spinal cord, so that there is an increasing discrepancy between the anatomical level of spinal cord segments and their corresponding vertebrae ([p. 252](#)). At stage 23, the vertebral column and spinal cord are the same length and the cord ends at the last coccygeal vertebra. In later fetal life, the conus medullaris lies between the third lumbar and fifth sacral vertebrae. In premature and term neonates, it lies between the first and third lumbar vertebrae, and in children between the ages of 1 and 7 years, it lies between the twelfth thoracic and third lumbar vertebrae ([Barson 1970](#), [Vettivel 1991](#), [Malas et al 2001](#), [Kesler et al 2007](#), [Suresh et al 2013](#)). In the adult, the spinal cord terminates on average at the level of the middle third of the body of the first lumbar vertebra ([Macdonald et al 1999](#)) (see [Fig. 43.79](#)), which corresponds approximately to the transpyloric plane. However, it may end as high as the middle third of the body of the eleventh thoracic vertebra or as low as the middle third of the body of the third lumbar vertebra ([Macdonald et al 1999](#)). Its position rises slightly in vertebral flexion, and there is some correlation with the length of the trunk, especially in females. (For dimensional data, consult [Barson and Sands \(1977\)](#).)

The filum terminale, a filament of connective tissue approximately 20 cm long, descends from the apex of the conus medullaris. Its upper 15 cm, the filum terminale internum, is continued within extensions of the dural and arachnoid meninges and reaches the caudal border of the second sacral vertebra. Its final 5 cm, the filum terminale externum, fuses with the investing dura mater, and then descends to the dorsum of the first coccygeal vertebral segment. The filum is continuous above with the spinal pia mater. A few strands of nerve fibres, which probably represent the roots of rudimentary second and third coccygeal spinal nerves, adhere to its upper part. The central canal is continued into the filum for 5–6 mm. A capacious part of the subarachnoid space surrounds the filum terminale internum, and is the site of election for access to the cerebrospinal fluid (CSF) via a lumbar puncture (see below).

The spinal cord varies in transverse width, gradually tapering cranio-caudally, except at the levels of the enlargements. It is not cylindrical, being wider transversely at all levels, especially in the cervical segments. The cervical enlargement, the source of the large spinal nerves that supply the upper limbs, extends from the third cervical to the second thoracic segments. Its maximum circumference (approximately 38 mm) is in the sixth cervical segment (a spinal cord segment provides the attachment of the rootlets of a pair of spinal nerves). The lumbar enlargement, the source of the large spinal nerves that supply the lower limbs, extends from the first lumbar to the third sacral segments; the equivalent vertebral levels are the ninth to twelfth thoracic vertebrae. Its greatest circumference (approximately 35 mm) is near the lower part of the body of the twelfth thoracic vertebra, below which it rapidly dwindles into the conus medullaris.

Fissures and sulci extend along most of the external surface. A ventral (anterior) median fissure (sulcus) and a posterior median sulcus and septum almost completely separate the cord into right and left halves; they are joined by a commissural band of nervous tissue that contains a central canal (see [Fig. 20.1](#)).

The ventral median fissure extends along the whole ventral surface with an average depth of 3 mm, although it is deeper at caudal levels. It contains a reticulum of pia mater. Dorsal to it is the anterior white commissure. Perforating branches of the spinal vessels pass from the fissure to the commissure to supply the central spinal region. The posterior median sulcus is shallower, and from it a posterior median septum penetrates more than halfway into the cord, almost to the central canal. The septum varies in anteroposterior extent from 4 to 6 mm, and diminishes caudally as the canal becomes more dorsally placed and the cord contracts. The patency of the central canal diminishes with age: a postmortem study found the canal to be patent along the length of the cord in infants under 1 year of age, but occluded in most segments with increasing age after the second decade ([Yasui et al 1999](#)).

Dorsal rootlets of spinal nerves enter the cord along a posterolateral sulcus that lies from 1.5 to 2.5 mm lateral to each side of the posterior median sulcus. The white matter between the posteromedian and posterolateral sulci on each side is the dorsal (posterior) funiculus. In cervical and upper thoracic segments, a longitudinal posterointermediate sulcus marks a septum dividing each posterior funiculus into two large tracts on either side, a medial fasciculus gracilis and a lateral fasciculus cuneatus. A ventrolateral (anterolateral) funiculus lies between the posterolateral sulcus and ventral median fissure, and is subdivided into ventral (anterior) and lateral funiculi by ventral spinal rootlets that pass through its substance to issue from the surface of the cord. The ventral funiculus is medial to, and includes, the emerging ventral rootlets, whilst the lateral funiculus lies between the roots and the posterolateral sulcus. In upper cervical segments, nerve rootlets emerge through each lateral funiculus to form the accessory nerve, which ascends in the vertebral canal lateral to the spinal cord and enters the posterior cranial fossa via the foramen magnum (see [Fig. 29.11](#)).

DORSAL AND VENTRAL ROOTS

The paired dorsal and ventral roots of the spinal nerves are continuous with the spinal cord (see [Fig. 45.1F](#)). They cross the subarachnoid space and traverse the dura mater separately, uniting in or close to their intervertebral foramina to form the (mixed) spinal nerves. Since the spinal cord is shorter than the vertebral column, the more caudal spinal roots descend for varying distances around and beyond the cord to reach their corresponding foramina. In so doing, they form a divergent sheaf of spinal nerve roots, the cauda equina, which is gathered round the filum terminale in the spinal theca, mostly distal to the apex of the cord.

Ventral spinal roots contain efferent somatic and, at some levels, preganglionic sympathetic, axons that extend from neuronal cell bodies

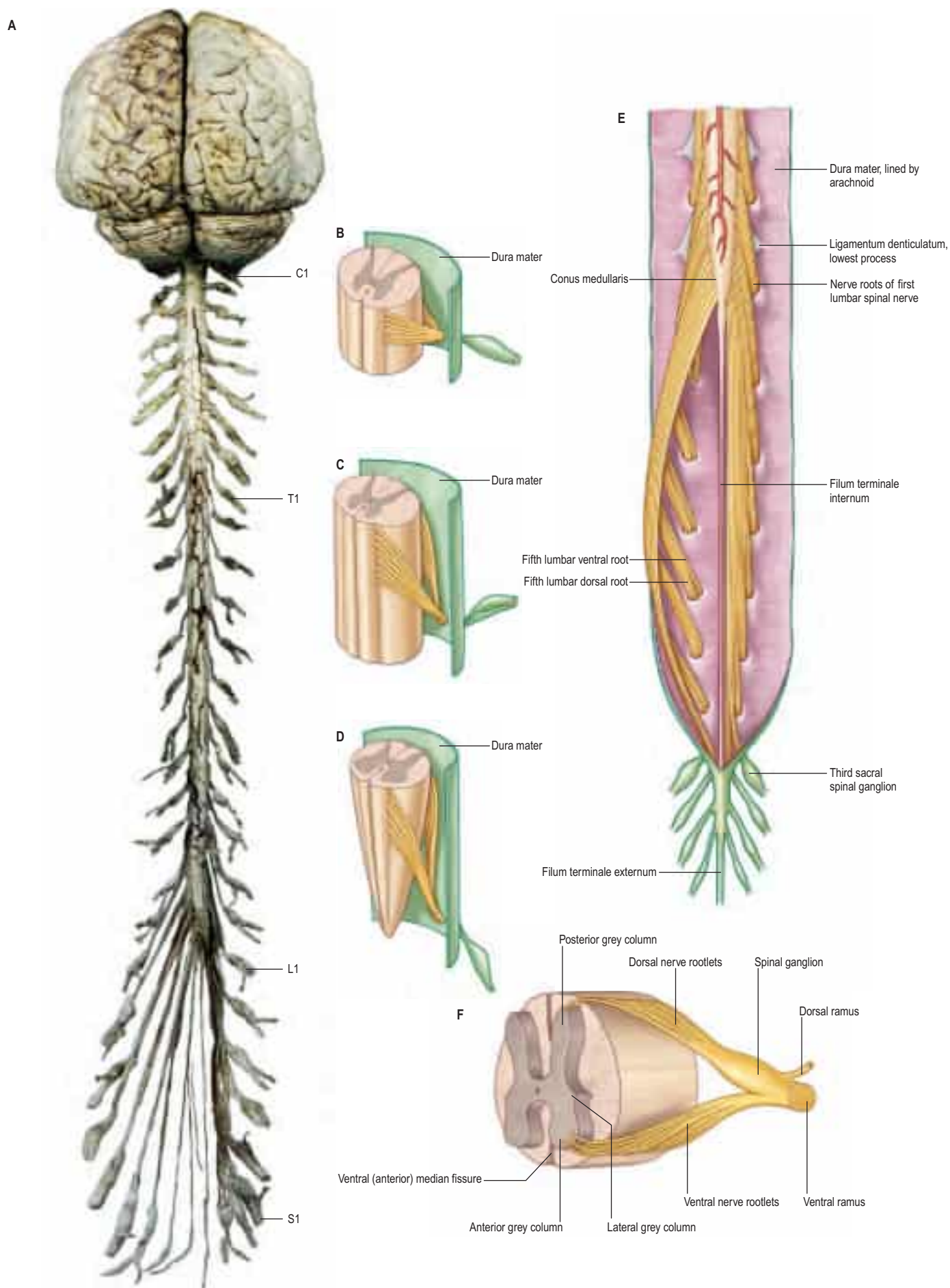


Fig. 45.1 **A**, The brain and spinal cord with attached spinal nerve roots and dorsal root ganglia, photographed from the dorsal aspect. Note the fusiform cervical and lumbar enlargements of the cord, and the changing obliquity of the spinal nerve roots as the cord is descended. The cauda equina is undisturbed on the right but has been spread out on the left to show its individual components. **B–D**, The formation of a typical spinal nerve, ventral aspect. **B**, Cervical level. **C**, Thoracic level. **D**, Lumbar level. **E**, The lower end of the spinal cord, filum terminale and cauda equina exposed from behind. The dura mater and the arachnoid have been opened and spread out. **F**, A spinal cord segment showing the mode of formation of a typical spinal nerve and the gross relationships of the grey and white matter. (A, Dissection by MCE Hutchinsonson, photograph by Kevin Fitzpatrick, London. B–D, With permission from Waschke J, Paulsen F (eds), Sobotta Atlas of Human Anatomy, 15th ed, Elsevier, Urban & Fischer. Copyright 2013.)

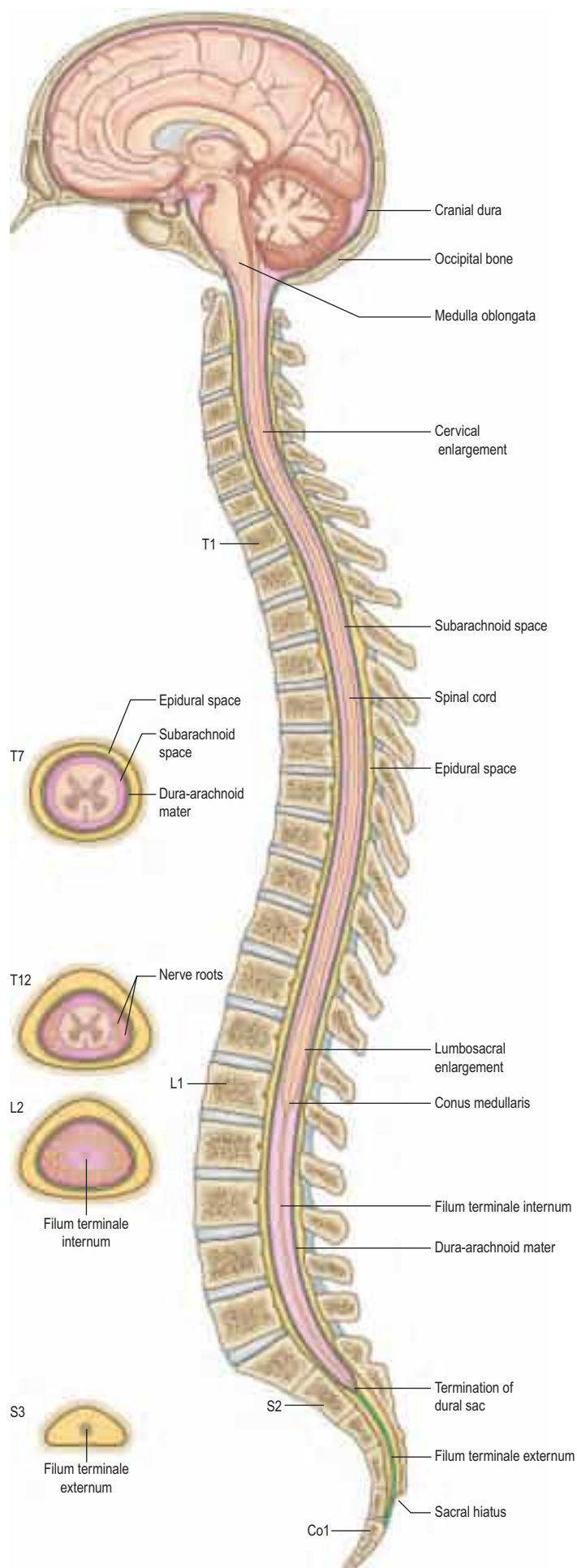


Fig. 45.2 The epidural and subarachnoid spaces.

in the ventral horns and intermediolateral columns, respectively. There are also afferent nerve fibres in these roots. The rootlets comprising each ventral root emerge from the anterolateral sulcus in groups over an elongated vertical elliptical area (see Fig. 45.1F). Dorsal spinal roots bear ovoid swellings, the spinal ganglia, one on each root proximal to its junction with a corresponding ventral root in an intervertebral foramen. Each root fans out into 6–8 rootlets before entering the cord in a vertical row in the posterolateral sulcus. Dorsal roots are usually said to contain only afferent axons (both somatic and visceral), which are the central processes of unipolar neurones in the spinal root ganglia, but they may also contain a small number (3%) of efferent fibres and autonomic vasodilator fibres.

Each ganglionic neurone has a single short stem that divides into a medial (central) branch that enters the spinal cord via a dorsal root, and a lateral (peripheral) branch that passes peripherally to a sensory end organ. The central branch is an axon while the peripheral one is an elongated dendrite (but when traversing a peripheral nerve is, in general structural terms, indistinguishable from an axon). The region of spinal cord associated with the emergence of a pair of nerves is a spinal segment but there is no actual surface indication of segmentation. Moreover, the deep neural sources or destinations of radicular fibres may lie far beyond the confines of the 'segment' so defined.

MENINGES

DURA MATER

In some areas within the skull, the dura mater can be distinguished from the endosteum, but at the base of the skull around the foramen magnum the two layers are fused and adherent to the bone. Distal to the foramen magnum, within the vertebral column, the dura is distinct from the tissues that line the vertebral canal, and separated from them by the epidural space (see below). The spinal dura mater forms a tube whose upper end is attached to the edge of the foramen magnum and to the posterior surfaces of the second and third cervical vertebral bodies, and also by fibrous bands to the posterior longitudinal ligament, especially towards the caudal end of the vertebral canal. The dural tube narrows at the lower border of the second sacral vertebra. It invests the thin spinal filum terminale, descends to the back of the coccyx, and blends with the periosteum.

Epidural space

The epidural space lies between the spinal dura mater and the tissues that line the vertebral canal (see Fig. 45.2). It is closed above by fusion of the spinal dura with the edge of the foramen magnum, and below by the posterior sacrococcygeal ligament that closes the sacral hiatus. It contains loosely packed connective tissue, fat, a venous plexus, small arterial branches, lymphatics and fine fibrous bands that connect the theca with the lining tissue of the vertebral canal. These bands, the meningovertbral ligaments, are best developed anteriorly and laterally. Similar bands tether the nerve root sheaths or 'sleeves' within their canals. There is also a midline attachment from the posterior spinal dura to the ligamentum nuchae at atlanto-occipital and atlanto-axial levels (Dean and Mitchell 2002). The venous plexus consists of longitudinally arranged chains of vessels, connected by circumdural venous 'rings'. The anteriorly placed vessels receive the basivertebral veins.

The shape of the space within each spinal segment is not uniform, though the segmental pattern is metamerically repeated. It is difficult to define the true shape of the 'space' because it changes with the introduction of fluid or as a result of preservation techniques. In the lumbar region, the dura mater is apposed to the walls of the vertebral canal anteriorly and attached by connective tissue in a manner that permits displacement of the dural sac during movement and venous engorgement. Adipose tissue is present posteriorly in recesses between the ligamentum flavum and the dura. The connective tissue extends for a short distance through the intervertebral foramina along the sheaths of the spinal nerves. Like the main thecal sac, the root sheaths are partially tethered to the walls of the foramina by fine meningovertbral ligaments.

Epidural injections

Contrast media and other fluids injected into the epidural space at the sacral level can spread up to the cranial base. Local anaesthetics injected near the spinal nerves, just outside the intervertebral foramina, may spread up or down the epidural space to affect the adjacent spinal nerves or may pass to the opposite side. The paravertebral spaces of

A variety of pathological processes can occur within the epidural space, compressing the dura and resulting in pain and potential neurological disturbance. Among the more common pathological entities seen are infection, haematoma and tumours (Figs 45.3–45.4).



Fig. 45.3 A T1 post-contrast sagittal lumbar spinal magnetic resonance (MR) image showing a spinal epidural abscess (arrow) adherent to the lumbar dura and spinal nerves in a 28-year-old female with a history of intravenous drug abuse.



Fig. 45.4 A sagittal thoracic MR image showing a large ventral spinal epidural haematoma (arrow) resulting in severe neck and upper thoracic pain and ataxia.

each side communicate via the epidural space, particularly at lumbar levels.

For a review of the morphology of the epidural space and a discussion of the nature of the lining layer of the vertebral canal, see [Newell \(1999\)](#).

Subdural space

The subdural space is a potential space in the normal spine because the arachnoid and dura are closely apposed ([Haines et al 1993](#)). It does not connect with the subarachnoid space but continues for a short distance along the cranial and spinal nerves. Accidental subdural catheterization may occur during epidural injections. Injection of fluid into the subdural space may damage the cord either by direct toxic effects or by compression of the vasculature.

ARACHNOID MATER

The spinal arachnoid mater surrounds the spinal cord and is continuous with the cranial arachnoid mater ([Fig. 45.5](#)). It is closely applied to the deep aspect of the dura mater. At sites where vessels and nerves enter or leave the subarachnoid space, the arachnoid mater is reflected on to the surface of these structures and forms a thin coating of leptomeningeal cells over the surface of both vessels and nerves. Thus a subarachnoid angle is formed as nerves pass through the dura into the intervertebral foramina. At this point, the layers of leptomeninges (arachnoid and pia) fuse and become continuous with the perineurium. The epineurium is in continuity with the dura. Such an arrangement seals the subarachnoid space so that particulate matter does not pass directly from the subarachnoid space into nerves. The existence of a pathway of lymphatic drainage from the CSF is controversial.

PIA MATER

The spinal pia mater (see [Fig. 45.5](#)) closely invests the surface of the spinal cord and passes into the ventral median fissure. The subpial collagenous layer in the spinal subpial 'space' is thicker than it is in the cerebral region and is continuous with the collagenous core of the ligamentum denticulatum (denticulate ligament).

The ligamentum denticulatum is a flat, fibrous sheet on either side of the spinal cord between the ventral and dorsal spinal roots. Its medial

border is continuous with the subpial connective tissue of the cord and its lateral border forms a series of triangular processes, the apices of which are fixed at intervals to the dura mater. The first crosses behind the vertebral artery where it is attached to the dura mater, and is separated by the artery from the first cervical ventral root. Its site of attachment to the dura mater is above the rim of the foramen magnum, just behind the hypoglossal nerve; the accessory nerve ascends on its posterior aspect (see [Fig. 29.11](#)). The last of the dentate ligaments lies between the exiting twelfth thoracic and first lumbar spinal nerves and is a narrow, oblique band that descends laterally from the conus medullaris. Changes in the form and position of the dentate ligaments during spinal movements have been demonstrated by cine-radiography. Beyond the conus medullaris, the pia mater continues as a coating of the filum terminale.

INTERMEDIATE LAYER

In addition to the well-defined coats of arachnoid and pia mater, the cord is also surrounded by an extensive intermediate layer of leptomeninges. This layer is concentrated in the dorsal and ventral regions, and forms a highly perforated, almost lace-like structure that is focally compacted to form the dorsal, dorsolateral and ventral ligaments of the spinal cord. Dorsally, the intermediate layer is adherent to the deep aspect of the arachnoid mater and forms a discontinuous series of dorsal ligaments that attach the spinal cord to the arachnoid. The dorsolateral ligaments are more delicate and fenestrated, and they extend from the dorsal roots to the parietal arachnoid. As the intermediate layer spreads laterally over the dorsal surface of the dorsal roots, it becomes increasingly perforated and eventually disappears. A similar arrangement is seen over the ventral aspect of the spinal cord but the intermediate layer is less substantial.

The intermediate layer is structurally similar to the trabeculae that cross the cranial subarachnoid space, i.e. it has a collagenous core coated by leptomeningeal cells. The intermediate layers of leptomeninges around the spinal cord may act as a baffle within the subarachnoid space to dampen waves of CSF movement in the vertebral canal. Inflammation within the spinal subarachnoid space may result in extensive fibrosis within the intermediate layer and the complications of chronic arachnoiditis ([Fig. 45.6](#)).

COVERINGS AND RELATIONS OF THE SPINAL ROOTS AND NERVES IN THE RADICULAR CANAL

Tubular prolongations of spinal dura mater, closely lined by arachnoid, extend around the spinal roots and nerves as they pass through the lateral zone of the vertebral canal and through the intervertebral foramina (see [Figs 45.5, 45.6A](#)). These prolongations, the spinal nerve sheaths or root sheaths, gradually lengthen as the spinal roots become increasingly oblique. Each individual dorsal and ventral root runs in the subarachnoid space with its own covering of pia mater. Each root pierces the dura separately, taking a sleeve of arachnoid with it, before joining within the dural prolongation just distal to the spinal ganglion. The dural sheaths of the spinal nerves fuse with the epineurium, within or slightly beyond the intervertebral foramina. The arachnoid prolongations within the sheaths do not extend as far distally as their dural coverings, but the subarachnoid space and the CSF it contains extend sufficiently distally to form a radiologically demonstrable root sleeve for each nerve. Shortening or obstruction of this sleeve seen on magnetic resonance imaging (MRI) indicates compression of the spinal nerve. At the cervical level, where the nerves are short and the vertebral movement is greatest, the dural sheaths are tethered to the periosteum of the adjacent transverse processes. In the lumbosacral region, there is less tethering of the dura to the periosteum, though there may be an attachment posteriorly to the facet joint capsule.

CEREBROSPINAL FLUID (CSF)

The cerebrospinal fluid is described in detail in [Chapter 18](#). Although there is free communication between the spinal and cerebral subarachnoid spaces, the mode of circulation of the spinal CSF and the contribution that it makes to the overall circulation of CSF remain uncertain in humans: CSF may be absorbed from the spinal subarachnoid space; spinal arachnoid granulations and villi have been described ([Kido et al 1976](#)).

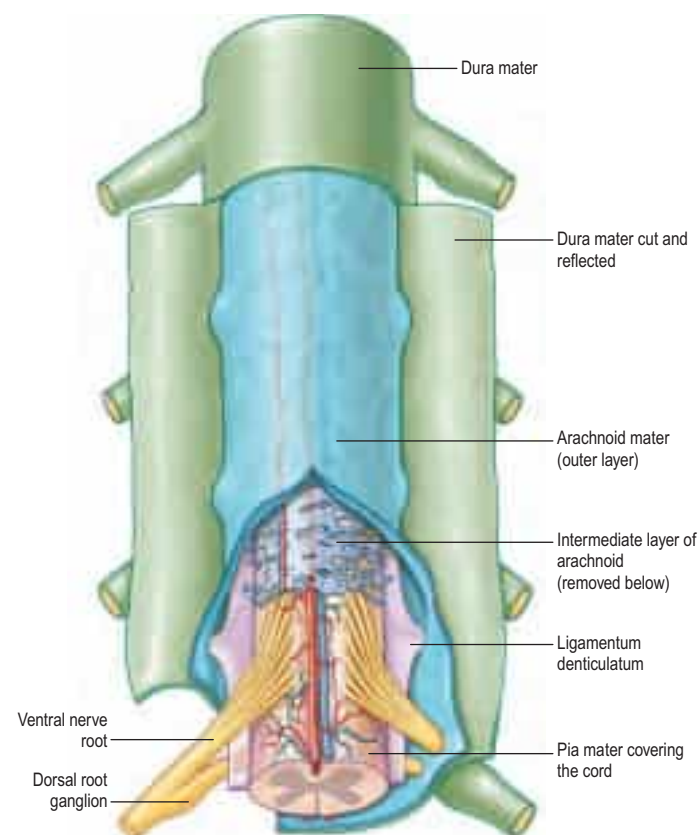


Fig. 45.5 Part of the spinal cord exposed from the anterior aspect to show the meningeal coverings.

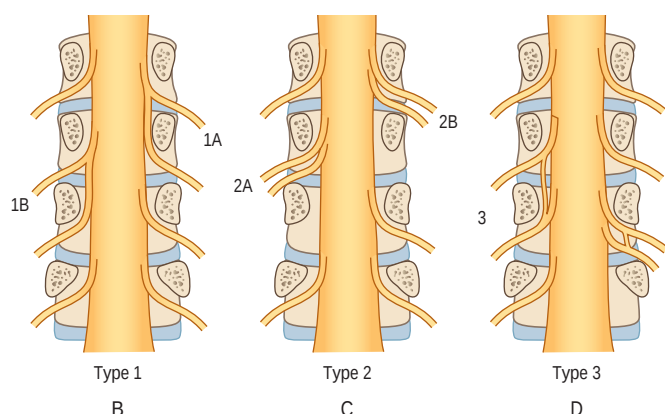
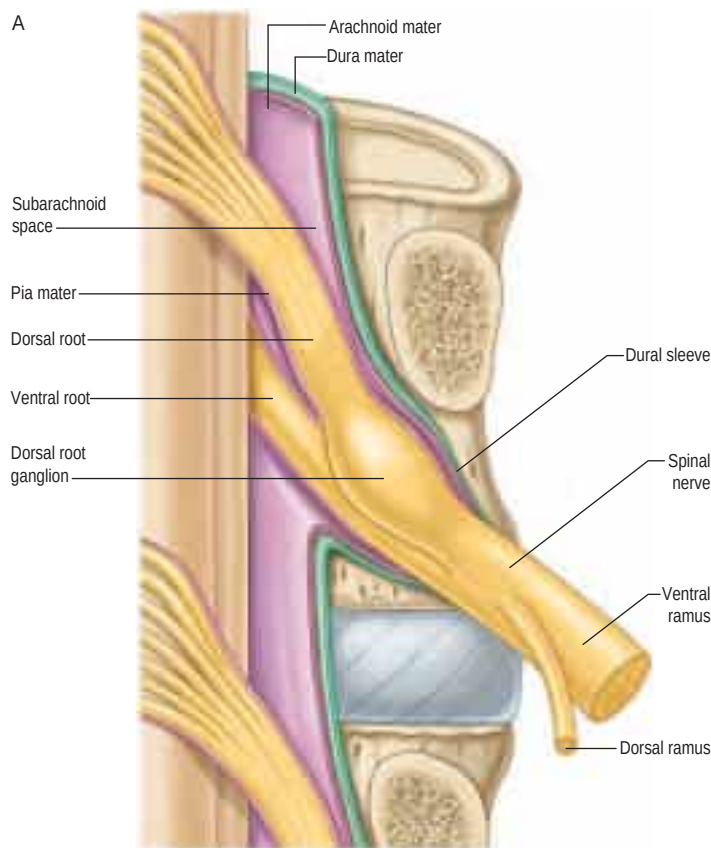


Fig. 45.6 A, A lumbar spinal nerve and its roots and meningeal coverings. B–D, Extradural anomalies of the lumbar nerve roots.

SPINAL NERVES

In those body segments that largely retain a metameric (segmental) structure, e.g. the thoracic region, spinal nerves show a common plan (Fig. 45.7). The dorsal, epaxial, ramus passes back lateral to the articular processes of the vertebrae and divides into medial and lateral branches that penetrate the deeper muscles of the back; both branches innervate the adjacent muscles and supply a band of skin from the posterior median line to the lateral border of the scapula (Fig. 45.8). The ventral, hypaxial, ramus is connected to a corresponding sympathetic ganglion by white and grey rami communicantes. It innervates the prevertebral muscles and curves around in the body wall to supply the lateral muscles of the trunk. Near the mid-axillary line, it gives off a lateral branch that pierces the muscles and divides into anterior and posterior cutaneous branches. The main nerve advances in the body wall, where it supplies the ventral muscles and terminates in branches to the skin.

Spinal nerves are united ventral and dorsal spinal roots, attached in series to the sides of the spinal cord. The term spinal nerve strictly applies only to the short segment after union of the roots and before branching occurs. This segment, the spinal nerve proper, lies in the intervertebral foramen; it is sometimes mistakenly called the 'nerve root'. There are 31 pairs of spinal nerves: 8 cervical, 12 thoracic, 5 lumbar, 5 sacral and 1 coccygeal. The abbreviations C, T, L, S and Co, with appropriate numerals, are commonly applied to individual nerves

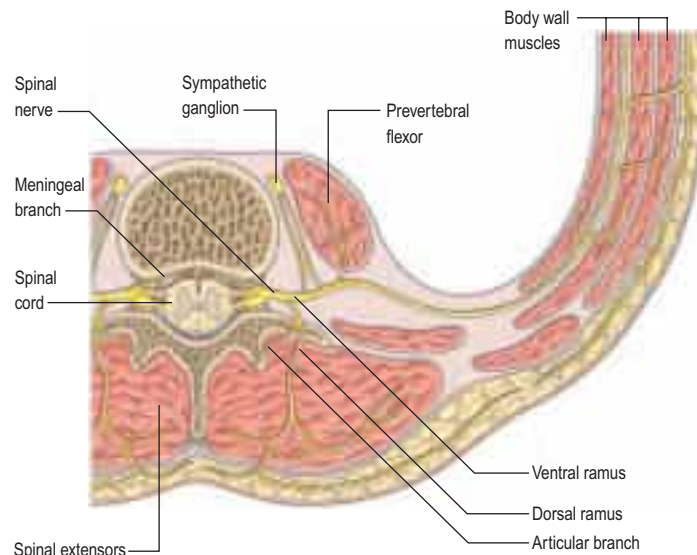


Fig. 45.7 The formation and branching pattern of a typical spinal nerve.

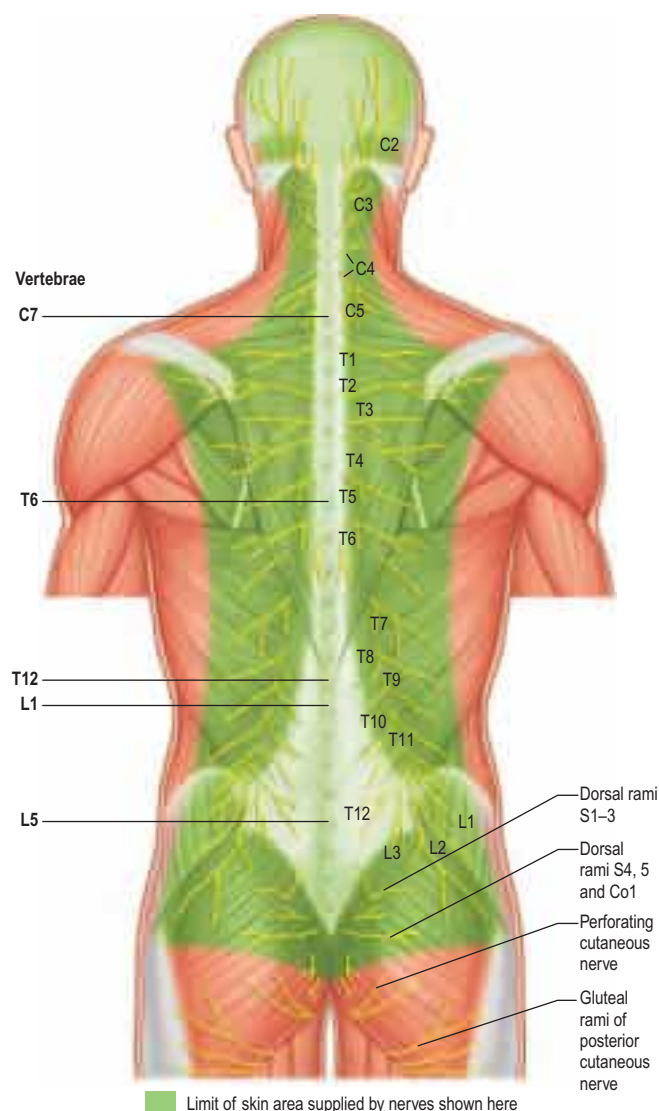


Fig. 45.8 The cutaneous distribution of the dorsal rami of the spinal nerves. The nerves are shown lying on the superficial muscles. The nerves are numbered on the right side; the spines of the seventh cervical, sixth and twelfth thoracic, and first and fifth lumbar vertebrae are labelled on the left side.

as individual 'root values'. The peripheral nerves emerge through the intervertebral foramina. At thoracic, lumbar, sacral and coccygeal levels, the numbered nerve exits the vertebral canal by passing below the pedicle of the corresponding vertebra, e.g. L4 nerve exits the intervertebral foramen between L4 and L5. However, in the cervical region, nerves

C1–7 pass above their corresponding vertebrae. C1 leaves the vertebral canal between the occipital bone and atlas, and hence is often termed the suboccipital nerve. The last pair of cervical nerves does not have a correspondingly numbered vertebra and C8 passes between the seventh cervical and first thoracic vertebrae. Each nerve is continuous with the spinal cord by ventral and dorsal roots; each dorsal root bears a spinal, sensory ganglion (dorsal root ganglion).

SPINAL ROOTS AND GANGLIA

Ventral (anterior) roots

Ventral roots contain axons of neurones in the ventral (anterior) and lateral spinal grey columns. Each emerges as a series of rootlets in two or three irregular rows in an area approximately 3 mm in horizontal width.

Dorsal (posterior) roots

Dorsal roots contain centripetal processes of neurones sited in the spinal ganglia. Each consists of medial and lateral fascicles that both diverge into rootlets that enter the spinal cord along the posterolateral sulcus. The rootlets of adjacent dorsal roots are often connected by oblique filaments, especially in the lower cervical and lumbosacral regions.

Little is known of the detail of the regions of entry and emergence of afferent and efferent rootlets in humans, but these zones of transition between the central and peripheral nervous systems have been extensively described in rodents (Fraher 2000).

Appearance and orientation of roots at each spinal level

The size and direction of spinal nerve roots vary. The upper four cervical roots are small, while the lower four are large. Cervical dorsal roots have a thickness ratio to the ventral roots of 3:1, which is greater than in other regions. The first dorsal root is an exception, being smaller than the ventral, and it is occasionally absent. The conventional view is that the first and second cervical spinal roots are short, running almost horizontally to their exits from the vertebral canal, and that from the third to the eighth cervical levels the roots slope obliquely down. Obliquity and length increase successively, although the distance between spinal attachment and vertebral exit never exceeds the height of one vertebra. An alternative view states that upper cervical roots descend, the fifth is horizontal, the sixth to eighth ascend, the first two thoracic roots are horizontal, the next three ascend, the sixth is horizontal and the rest descend (Kubik and Müntener 1969). This view is based on the observation that the cervicothoracic part of the spinal cord grows more in length than other parts.

Thoracic roots, except the first, are small, and the dorsal root only slightly exceeds the ventral in thickness. They increase successively in length. In the lower thoracic region, the roots descend in contact with the spinal cord for at least two vertebrae before emerging from the vertebral canal.

Lower lumbar and upper sacral roots are the largest, and their rootlets are the most numerous. Coccygeal roots are the smallest. Kubik and Müntener (1969) confirm that lumbar, sacral and coccygeal roots descend with increasing obliquity to their exits. The spinal cord ends near the lower border of the first lumbar vertebra, and so the lengths of successive roots rapidly increase; the consequent collection of roots is the cauda equina (see Fig. 45.1A). The largest roots, and hence the largest spinal nerves, are continuous with the spinal cervical and lumbar enlargements and innervate the upper and lower limbs.

Spinal ganglia (dorsal root ganglia)

Spinal ganglia are large groups of neurones on the dorsal spinal roots. Each is oval and reddish; its size is related to that of its root. A ganglion is bifid medially where the two fascicles of the dorsal root emerge to enter the cord. Ganglia are usually sited in the intervertebral foramina, immediately lateral to the perforation of the dura mater by the roots (see Fig. 45.1B). However, the first cervical ganglion lies on the vertebral arch of the atlas, the second lies behind the lateral atlanto-axial joint, the sacral lie inside the vertebral canal, and the coccygeal ganglion usually lies within the dura mater. The first cervical ganglia may be absent. Small aberrant ganglia sometimes occur on the upper cervical dorsal roots between the spinal ganglia and the cord.

SPINAL NERVES PROPER

Immediately distal to the spinal ganglia, ventral and dorsal roots unite to form spinal nerves. These very soon divide into dorsal and ventral rami, both of which receive fibres from both roots (see Fig. 45.6A). At all levels above the sacral, this division occurs within the intervertebral foramen. Division of the sacral spinal nerves occurs within the sacral vertebral canal, and the dorsal and ventral rami exit separately through posterior and anterior sacral foramina at each level. Spinal nerves trifurcate at some cervical and thoracic levels, in which case the third branch is called a ramus intermedius. At or distal to its origin, each ventral ramus gives off recurrent meningeal (sinuvertebral) branches and receives a grey ramus communicans from the corresponding sympathetic ganglion. The thoracic and first and second lumbar ventral rami each contribute a white ramus communicans to the corresponding sympathetic ganglia. The second, third and fourth sacral nerves also supply visceral branches, unconnected with sympathetic ganglia, which carry a parasympathetic outflow direct to the pelvic plexuses.

Cervical spinal nerves enlarge from the first to the sixth nerve. The seventh and eighth cervical and the first thoracic nerve are similar in size to the sixth cervical nerve. The remaining thoracic nerves are relatively small. Lumbar nerves are large, increasing in size from the first to the fifth. The first sacral is the largest spinal nerve; thereafter the sacral nerves decrease in size. The coccygeal nerves are the smallest spinal nerves. The size of the spinal nerve and its associated structures within the intervertebral foramen is not in direct relation to the size of the foramen. At lumbar levels, though L5 is the largest nerve, its foramen is smaller than those of L1–4, which renders this nerve particularly liable to compression.

In the radicular ('root') canal and intervertebral foramen, the spinal nerve is related to the spinal artery of that level and its radicular branch, and to a small plexus of veins. At the outer end of the foramen, the nerve may lie above or below transforaminal ligaments.

Meningeal nerves

Recurrent meningeal (or sinuvertebral) nerves (Fig. 45.9) occur at all vertebral levels. They are mixed sensory and sympathetic nerves, represented by numerous fine filaments amongst which one, or two to four, larger trunks may be evident. At cervical levels, the autonomic roots arise from the grey rami that form the vertebral nerve. At thoracic and lumbar levels, each nerve is formed by a somatic root from the ventral ramus and by an autonomic root from the grey ramus communicans of that segment. Each nerve pursues a recurrent course through the

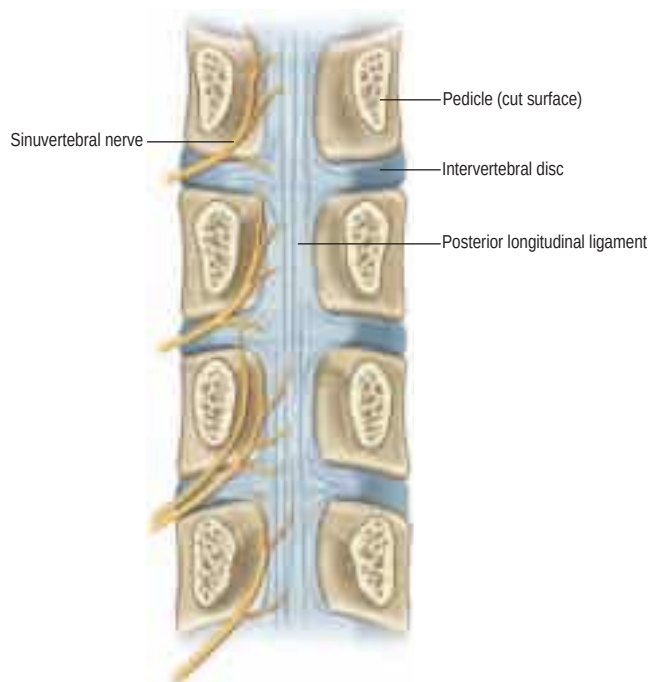


Fig. 45.9 The course and skeletal distribution of the lumbar sinuvertebral nerves. Each nerve supplies the intervertebral disc at its level of entry into the vertebral canal, the disc above, and the intervening posterior longitudinal ligament. In about one-third of cases, the nerve at a particular level may be represented by more than one filament.

intervertebral foramen, passing ventral to the spinal nerve, to enter the vertebral canal, where it divides into ascending, descending and transverse branches. These branches communicate with corresponding branches from the segments above and below, and from the opposite side, forming arcades along the floor of the vertebral canal. Meningeal branches of the arcades form a plexus on the ventral surface of the dural sac and nerve root sleeves that attenuates laterally; the posterior paramedian dura is devoid of nerve endings. Skeletal branches are distributed to the posterior longitudinal ligament, the periosteum of the vertebral bodies, and to the posterior and posterolateral aspects of the intervertebral discs (Garcia-Cosamalon et al 2010). Vascular branches accompany the veins and arteries of the vertebral canal and those of the vertebral bodies. The upper three cervical meningeal nerves ascend through the foramen magnum into the posterior cranial fossa, where they innervate the dura mater that covers the clivus. *En route*, they innervate the median atlanto-axial joint and its ligaments.

Functional components of spinal nerves

A typical spinal nerve contains somatic efferent fibres and somatic and visceral afferent fibres. Some, but not all, spinal nerves also contain preganglionic autonomic fibres.

Somatic components

Somatic efferent fibres innervate skeletal muscles and are axons of α , β and γ neurones in the spinal ventral grey column. Somatic afferent fibres convey impulses into the central nervous system from receptors in the skin, subcutaneous tissue, muscles, tendons, fasciae and joints; they are peripheral processes of unipolar neurones in the spinal ganglia.

Visceral components

Preganglionic visceral efferent sympathetic fibres are axons of neurones in the spinal intermediolateral grey column throughout the thoracic and upper two or three lumbar segments; they join the sympathetic trunk via corresponding white rami communicantes and synapse with postganglionic neurones that are distributed to smooth muscle, myocardium or exocrine glands. The preganglionic visceral efferent parasympathetic fibres are axons of neurones in the spinal lateral grey column of the second to fourth sacral segments; they leave the ventral rami of corresponding sacral nerves and synapse in pelvic ganglia. The postganglionic axons are distributed mainly to smooth muscle or glands in the walls of the pelvic viscera. Visceral afferent fibres have cell bodies in the spinal ganglia. Their peripheral processes pass through white rami communicantes and, without synapsing, through one or more sympathetic ganglia to end in the walls of the viscera. Some visceral afferent fibres may enter the spinal cord in the ventral roots.

Central processes of ganglionic unipolar neurones enter the spinal cord by dorsal roots and synapse on somatic or sympathetic efferent neurones, usually through interneurones, completing reflex paths. Alternatively, they may synapse with other neurones in the spinal or brainstem grey matter that give origin to a variety of ascending tracts.

VARIATIONS OF SPINAL ROOTS AND NERVES

The courses of spinal roots and nerves in relation to the thecal sac and vertebral and radicular canals may be aberrant. An individual intervertebral foramen may contain a duplicated sheath, nerve and roots, which will then be absent at an adjacent level. Abnormal communications between roots may occur within the vertebral canal. These anomalies have been described and classified for the lumbosacral spine by Neidre and Macnab (1983) (see Fig. 45.6).

RAMI OF THE SPINAL NERVES

Ventral (anterior) primary rami supply the limbs and the anterolateral aspects of the trunk, and in general are larger than the dorsal rami. Thoracic ventral rami run independently and retain a largely segmental distribution. Cervical, lumbar and sacral ventral rami connect near their origins to form plexuses. Dorsal rami do not join these plexuses. The ventral rami are described in the appropriate regional sections.

Dorsal (posterior) primary rami of spinal nerves are usually smaller than the ventral rami and are directed posteriorly. Retaining a segmental distribution, all, except for the first cervical, fourth and fifth sacral and the coccygeal, divide into medial and lateral branches that supply the muscles and skin of the posterior regions of the neck and trunk (see Fig. 45.8).

Cervical dorsal spinal rami

Each cervical spinal dorsal ramus, except the first, divides into medial and lateral branches that all innervate muscles. In general, only medial branches of the second to fourth, and usually the fifth, supply the skin. Except for the first and second, each dorsal ramus passes posteriorly around an articular pillar as far as the root of the transverse process medial to a posterior intertransverse muscle, which it supplies.

First cervical dorsal ramus (suboccipital nerve)

The first cervical dorsal ramus, the suboccipital nerve, is larger than the ventral (see Fig. 43.76). It emerges superior to the posterior arch of the atlas and inferior to the vertebral artery, and enters the suboccipital triangle to supply rectus capitis posterior major and minor, obliquus capitis superior and inferior, and semispinalis capitis. A filament from the branch to inferior oblique joins the second dorsal ramus. The suboccipital nerve occasionally has a cutaneous branch that accompanies the occipital artery to the scalp, and connects with the greater and lesser occipital nerves. It may also communicate with the accessory nerve.

Second cervical dorsal ramus

The second cervical dorsal ramus is slightly larger than the ventral and all the other cervical dorsal rami (see Figs 43.76, 30.22). It runs between the lamina of the axis and inferior oblique, below which it divides into a large medial and smaller lateral branch. The dorsal ramus or its medial branch receives communicating branches from the first cervical dorsal ramus that pass both through and around inferior oblique. A descending communicating branch crosses the C2–3 facet joint to reach the third cervical dorsal ramus.

Greater occipital nerve

The medial branch, termed the greater occipital nerve, passes transversely across inferior oblique deep to semispinalis capitis, which it innervates. Near the origin of inferior oblique, it turns rostrally along rectus capitis posterior major, where it receives a communicating branch from the third occipital nerve before turning dorsally to pierce semispinalis capitis. It emerges on to the scalp by passing above an aponeurotic sling between trapezius and sternocleidomastoid near their occipital attachments. It ascends with the occipital artery and divides into branches that connect with the third occipital and lesser occipital nerves, and which supply the back of the auricle and the skin of the scalp as far forwards as the coronal suture. The lateral branch of the second cervical dorsal ramus supplies longissimus capitis, semispinalis capitis and splenius capitis, and sends a communicating branch to the lateral branch of the third cervical dorsal ramus.

Greater occipital neuralgia

Greater occipital neuralgia is a syndrome of pain and paraesthesiae felt in the distribution of the greater occipital nerve. This is usually caused by damage or irritation to the greater occipital nerve but more rarely can result from a similar process in the lesser occipital nerve or both greater and lesser occipital nerves. A similar syndrome may be caused by upper facet joint arthritis involving the second cervical root (Vanel-deren et al 2010).

Third cervical dorsal ramus

The third cervical dorsal ramus is intermediate in size between the second and fourth. It courses back round the articular pillar of the third cervical vertebra, medial to the posterior intertransverse muscle, and divides into medial and lateral branches. The lateral branch passes dorsally across the surface of semispinalis capitis to supply it and the overlying longissimus capitis and splenius capitis. It receives a communicating branch from the lateral branch of the second cervical dorsal ramus, and sends a similar branch to the lateral branch of the fourth cervical dorsal ramus. The deep medial branch curves dorsally and medially around the waist of the articular pillar before ending in multifidus. It sends an articular branch to the C3–4 facet joint, and may also send a branch into semispinalis capitis. The superficial medial branch, the third occipital nerve, curves around the lateral and dorsal surfaces of the C2–3 facet joint, which it supplies. It continues transversely, deep to semispinalis, which it supplies, and sends a communicating branch to the greater occipital nerve. Just above the second cervical spinal process, the third occipital nerve turns dorsally to pierce semispinalis capitis, splenius capitis and trapezius, and becomes cutaneous over a small area immediately below the superior nuchal line. Communicating branches join the cutaneous branches of the greater and lesser occipital nerves.

Dorsal rami of the lower five cervical nerves

The dorsal rami of the lower five cervical nerves curve back round the vertebral articular pillars and divide into lateral and medial branches. The lateral branches supply longissimus cervicis, splenius cervicis and iliocostalis cervicis. The medial branches curve around the lateral and dorsal surfaces of the articular pillar of their segment, deep to semispinalis capitis. They send articular branches to the facet joints above and below their course before entering and supplying multifidus. Terminal branches reach interspinalis at each segment. Cutaneous branches arise consistently from the fourth, and commonly from the fifth, sixth and eighth, but not usually from the seventh medial branches. These branches pass across multifidus or through semispinalis cervicis and turn dorsally around the medial border of semispinalis capitis, to pierce splenius cervicis and trapezius and reach the skin.

Thoracic dorsal spinal rami

At each segmental level, the thoracic dorsal rami pass dorsally through an aperture bounded by the transverse processes superiorly and inferiorly, the facet joint medially, and the superior costotransverse ligament laterally. Thereafter, each dorsal ramus follows a prolonged course laterally, between the anterior lamella of the superior costotransverse ligament anteriorly, and the costolamellar ligament and the posterior lamella of the superior costotransverse ligament posteriorly. In this narrow space, the dorsal ramus divides into lateral and medial branches. The lateral branches continue laterally as far as the costotransverse joint, above which they turn dorsally and inferiorly to supply the levatores costarum, and enter the longissimus thoracis and iliocostalis muscles. Lateral branches from the upper six thoracic dorsal rami remain intramuscular. Those from the lower six levels emerge from iliocostalis lumborum to pierce serratus posterior inferior and latissimus dorsi in line with the costal angles. They descend across as many as four ribs before becoming superficial. The twelfth thoracic lateral branch sends a filament medially along the iliac crest and then passes down to the anterior gluteal skin. Some upper thoracic lateral branches may become cutaneous.

Each medial branch hooks dorsally around the lateral margin of the posterior lamella of the superior costotransverse ligament, lying a variable distance above the tip of the transverse process. At the first to fourth, and at the ninth and tenth thoracic levels, the medial branches cross the superior tip of the transverse process to turn inferiorly and medially across the posterior surface of the transverse process, lying between the attachment sites of multifidus medially and semispinalis laterally. Each nerve continues this oblique course to supply multifidus and semispinalis. At the fifth to eighth thoracic levels, the medial branches assume a course parallel to that of other levels, but do not reach the transverse process, instead passing superior and dorsal to the plane of the transverse processes. At the eleventh and twelfth levels, in keeping with the relatively short transverse processes at these levels, the medial branches run close to the superior articular processes. Each medial branch supplies articular branches to the facet joints above and below its course. Medial branches of the upper six thoracic dorsal rami furnish cutaneous branches that pierce trapezius and the rhomboids to reach the skin near the spinous processes. Cutaneous branches may occasionally arise from the medial branches of the lower six thoracic dorsal rami.

Lumbar dorsal spinal rami

After leaving its spinal nerve, each of the first to fourth lumbar dorsal rami passes through an aperture in the dorsal leaf of the intertransverse ligament just above a transverse process (Fig. 45.10). Each dorsal ramus supplies the overlying medial intertransverse muscle before dividing into lateral and medial branches. The lateral branches supply the longissimus and iliocostalis components of the lumbar erector spinae. The upper three lumbar lateral branches emerge from the lateral border of iliocostalis lumborum and pierce the aponeurosis of latissimus dorsi to become cutaneous. They cross the iliac crest to reach the gluteal skin; some reach as far as the level of the greater trochanter. Medial branches cross the junction of the superior articular process and transverse process, and hook medially between the mammillary and accessory processes, deep to the mammillo-accessory ligament. Medial to the ligament, each medial branch sends articular branches to the facet joints above and below its course, before entering multifidus, which it supplies. Terminal branches reach interspinalis at each segment.

The fifth dorsal ramus has a longer course than other dorsal rami. It arches over the ala of the sacrum, lying against the root of the first sacral

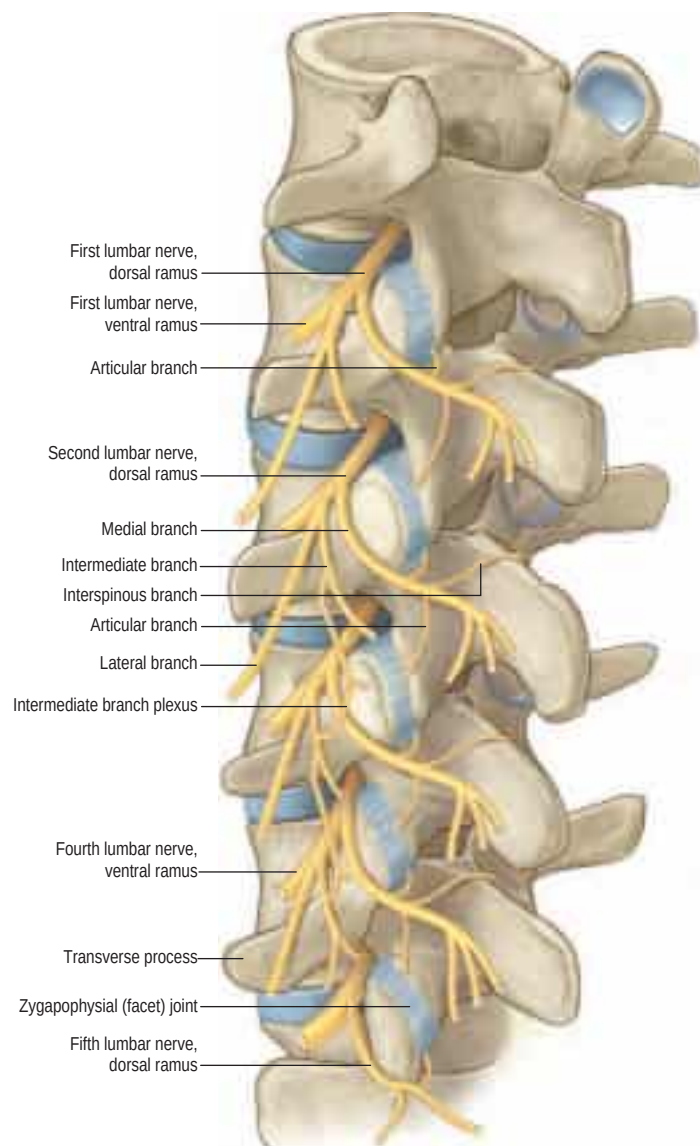


Fig. 45.10 A left posterolateral view of the lumbar spine showing the branches of the lumbar dorsal rami.

superior articular process. Opposite the base of the lumbosacral joint, it sends a branch into the lowest fibres of longissimus thoracis, and a communicating branch to the first sacral dorsal ramus, before terminating as a medial branch that hooks around the lumbosacral joint to end in multifidus.

Sacral dorsal spinal rami

Sacral dorsal rami are small, diminishing in size inferiorly; other than the fifth, all emerge through the dorsal sacral foramina. The upper three are covered at their exit by multifidus, and divide into medial and lateral branches. Medial branches are small and end in multifidus. Lateral branches join together and with branches of the last lumbar and fourth sacral dorsal rami to form loops dorsal to the sacrum. Branches from these loops run dorsal to the sacrotuberous ligament and form a second series of loops under gluteus maximus. From these, two or three gluteal branches pierce gluteus maximus (along a line from the posterior superior iliac spine to the coccygeal apex) to supply the posterior gluteal skin.

The dorsal rami of the fourth and fifth sacral nerves are small and lie caudal to multifidus. They unite with each other and with the coccygeal dorsal ramus to form loops dorsal to the sacrum; filaments from these supply the skin over the coccyx.

Coccygeal dorsal spinal ramus

The coccygeal dorsal spinal ramus does not divide into medial and lateral branches. It communicates with the fourth and fifth sacral dorsal rami to supply the skin over the coccyx.

VASCULAR SUPPLY OF SPINAL CORD, ROOTS AND NERVES

ARTERIES

The spinal cord and its roots and nerves are supplied with blood by both longitudinal and segmental vessels (Fig. 45.11). Three major longitudinal vessels, a single anterior and two posterior spinal arteries (each of which is sometimes doubled to pass on either side of the dorsal rootlets) originate intracranially from the vertebral artery and terminate in a plexus around the conus medullaris. The anterior spinal artery forms from the fused anterior spinal branches of the vertebral artery, and descends in the ventral median fissure of the cord. Each posterior spinal artery originates either directly from the ipsilateral vertebral artery or from its posterior inferior cerebellar branch, and descends in a posterolateral sulcus of the cord. The segmental arteries are derived in craniocaudal sequence from spinal branches of the vertebral, deep cervical, intercostal and lumbar arteries. These vessels enter the vertebral canal through the intervertebral foramina and anastomose with branches of the longitudinal vessels to form a pial plexus on the surface of the cord. The segmental spinal arteries send anterior and posterior radicular branches to the spinal cord along the ventral and dorsal roots. Most anterior radicular arteries are small and end in the ventral nerve roots or in the pial plexus of the cord. The small posterior radicular arteries also supply the dorsal root ganglia; branches enter at both ganglionic poles to be distributed around ganglion cells and nerve fibres. (See also Crock (1996).)

Segmental radiculomedullary feeder arteries

Some radicular arteries, mainly situated in the lower cervical, lower thoracic and upper lumbar regions, are large enough to reach the ventral median fissure, where they divide into slender ascending and large descending branches. These are the anterior radiculomedullary feeder arteries (Domisse 1975). They anastomose with the anterior spinal arteries to form a single or partly double longitudinal vessel of uneven calibre along the ventral median fissure. The largest anterior medullary feeder, the great anterior radiculomedullary artery of Adamkiewicz, varies in level, arising from a spinal branch of either one of the lower posterior intercostal arteries (T9–11), or of the subcostal artery (T12), or less frequently of the upper lumbar arteries (L1 and L2). It most often arises on the left side (Carmichael and Glowiczki 1999). Reaching the spinal cord, it sends a branch to the anterior spinal artery below and another to anastomose with the ramus of the posterior spinal artery, which lies anterior to the dorsal roots. It may be the main supply to the lower two-thirds of the cord. Central branches of the anterior spinal artery enter the ventral median fissure, and then turn right or left to supply the ventral grey column, the base of the dorsal grey column, including the dorsal nucleus, and the adjacent white matter (Fig. 45.12).



Each posterior spinal artery contributes to a pair of longitudinal anastomotic channels, anterior and posterior to the dorsal spinal roots. These are reinforced by posterior medullary feeders from the posterior radicular arteries. The latter are variable in number and size, but smaller, more numerous and more evenly distributed than the anterior medullary feeders. The anterior channel is joined by a ramus from the descending branch of the great anterior segmental medullary artery of Adamkiewicz. In all longitudinal spinal arteries the width of the lumen is uneven and complete interruptions may occur. At the conus medullaris they communicate by anastomotic loops. Anastomoses other than those between the pial or peripheral spinal arterial branches may be important, e.g. a posterior spinal series of anastomoses between rami of the dorsal divisions of segmental arteries near the spinous processes.

Intramedullary arteries

The central branches of the anterior spinal artery supply about two-thirds of the cross-sectional area of the cord. The rest of the dorsal grey and white columns and peripheral parts of the lateral and ventral white columns are supplied by numerous small radial vessels that branch from posterior spinal arteries and the pial plexus. In a microangiographic study of the human cervical spinal cord, up to six anterior, and eight posterior, radicular spinal arteries were described, and up to eight central branches arose from each centimetre of the anterior spinal artery (Turnbull et al 1966).

Spinal cord ischaemia

The spinal cord can rely for neither its transverse nor its longitudinal blood supply entirely on the longitudinal arteries. The anterior longitudinal artery and the intramedullary arteries are functional end-arteries, although overlap of territories of supply has been described (Backes and Nijenhuis 2008). Damage to the anterior longitudinal artery can result in loss of function of the anterior two-thirds of the cord (Fig. 45.13). The longitudinal arteries cannot supply the whole length of the cord, and the input of the segmental medullary feeder vessels is essential. This is especially true of the artery of Adamkiewicz, which may effectively carry the major supply for the lower cord; damage to this individual artery compromises perfusion of the distal cord and may be responsible for paraplegia following aortic bypass procedures. For many years, it was assumed that the anterior spinal artery system provided the dominant supply to the cord, but clinical evidence implies that the posterior spinal arteries may be as important as the anterior system in protecting the cord. The mid-thoracic cord, distant from the main anterior medullary feeders, is particularly liable to become ischaemic after periods of hypotension; T4–9 has been described as the critical vascular zone of the spinal cord, where interference with the circulation is most likely to result in paraplegia.



VEINS

Intramedullary veins within the substance of the spinal cord drain into a circumferential plexus of surface veins, the coronal plexus (venous plexus of the pia mater). Six tortuous longitudinal channels are usually identified within this plexus: anterior and posterior spinal veins (anterior and posterior median veins) and four others that run on either side of the ventral and dorsal nerve roots (Fig. 45.14). Only the anterior spinal vein, which drains the central grey matter, is consistently complete. It receives sulcal veins and small pial veins from the coronal plexus and is largest in calibre in the lumbosacral region. The posterior spinal vein is frequently variable, appearing plexiform in some segments and as a single channel in others. These vessels connect freely and drain superiorly into the cerebellar veins and cranial sinuses, and segmentally into medullary veins mainly. The segmental veins drain into the intervertebral veins and thence into the external vertebral venous plexuses (see Fig. 43.21), the caval and azygos systems.

Segmental veins

Anterior and posterior radiculomedullary veins run along some of the ventral and dorsal roots. Like the medullary feeder arteries, they are largest in the cervical and lumbar regions of the cord, but do not necessarily occur in the same segments as the medullary feeders. They drain the cord but not the roots themselves. Very small anterior and posterior radicular veins occur in most spinal segments, accompanying and draining the ventral and dorsal roots and some of the cord at the points of entry and exit of the rootlets; they usually drain into the intervertebral veins.

THE EFFECTS OF INJURY

SPINAL CORD INJURY AND VERTEBRAL COLUMN INJURY

Various clinical definitions have been described in the *International Standards Booklet for Neurological and Functional Classification of Spinal Cord Injury*, which was a landmark publication in standardizing the language of the neurological examination as it relates to spinal cord injury (Ditunno et al 1994). 'Tetraplegia' refers to impairment or loss of neural function secondary to a cervical injury. 'Paraplegia' refers to loss of function in the thoracic, lumbar or sacral segments of the cord. 'Dermatome' refers to the area of skin innervated by cutaneous axons in the distribution of a single spinal nerve. 'Myotome' refers to the muscle fibres innervated by the motor axons of a single spinal nerve. 'Neurological level' is the first level where motor and sensory function is normal on both sides of the body. 'Sensory level' refers to the most caudal level with normal sensation on both sides of the body. 'Motor level' refers to the most caudal segment with antigravity strength, providing that the next most rostral level is stronger. 'Complete injury' occurs when there is an absence of motor and sensory function below the motor and sensory levels and absence of function in the lowest sacral segment. 'Incomplete injury' refers to partial preservation of

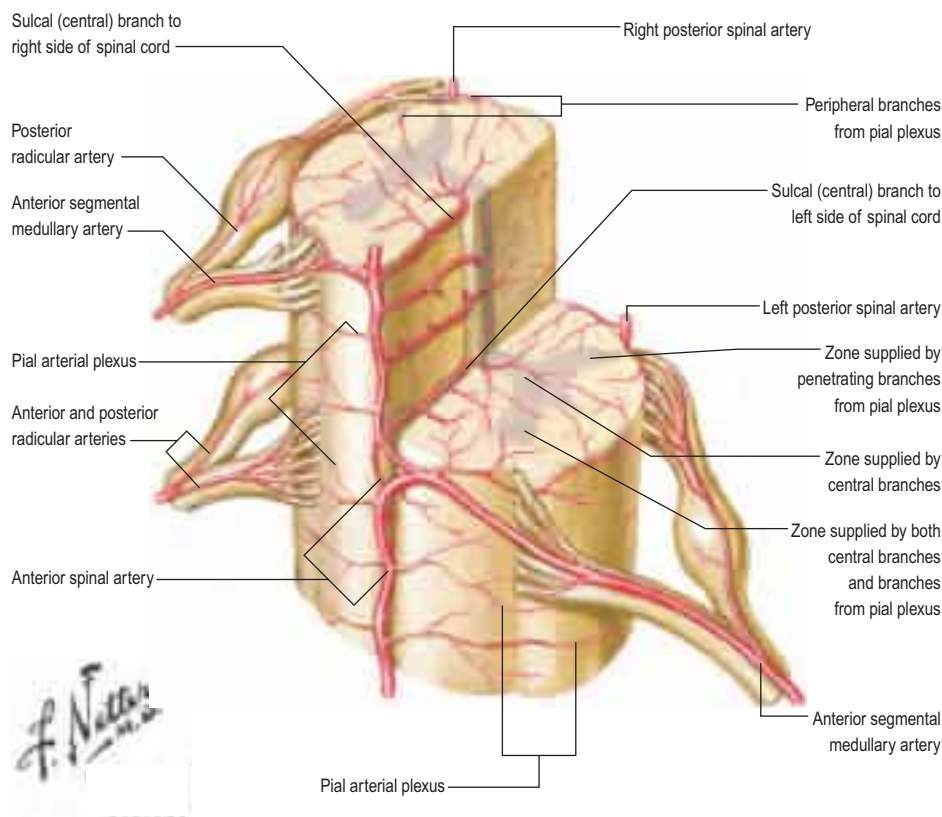


Fig. 45.12 Arterial disposition within the spinal cord. (Reprinted from Netter Anatomy Illustration Collection, © Elsevier Inc. All Rights Reserved.)



Fig. 45.13 A, A sagittal thoracic MR image showing spinal cord ischaemic changes (arrow) secondary to thrombotic occlusion of the artery of Adamkiewicz in a patient with a hypercoagulable disorder. **B**, The changes were noted to be in the distribution of the anterior spinal artery, as seen on this axial image.

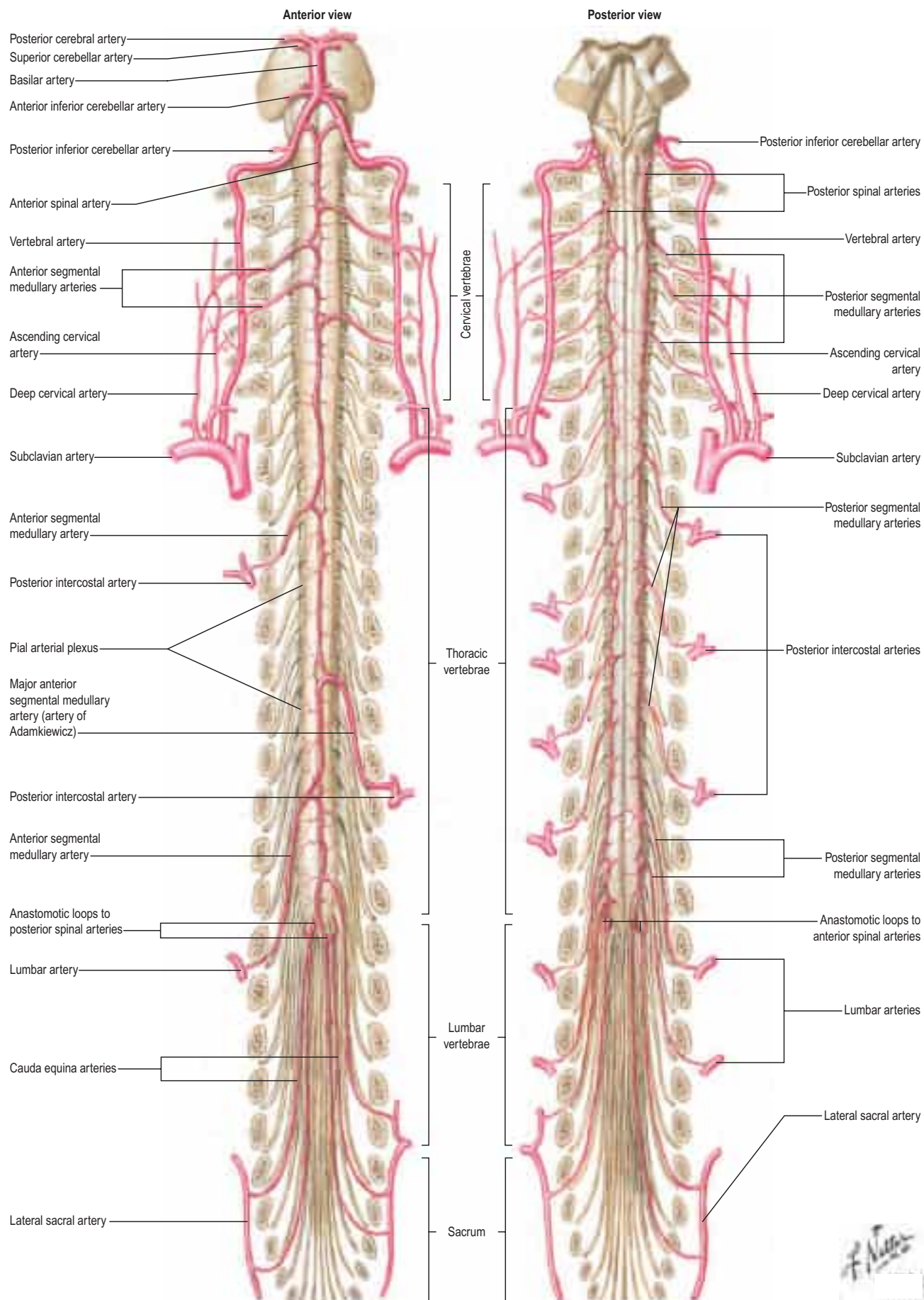


Fig. 45.11 Arteries of the spinal cord. (Reprinted from Netter Anatomy Illustration Collection, © Elsevier Inc. All Rights Reserved.)

sensory and/or motor function below the neurological levels, as well as evidence of 'sacral sparing' (evidence of preserved function of the lowest sacral segment) (Ditunno et al 2004). 'Spinal shock' refers to the absence of reflexes and motor/sensory function below the level of injury and occurs typically from the time at which spinal cord injury is

sustained to 24 hours after injury, followed by a phase of reflex return (Ditunno et al 2004).

In the assessment of a patient with spinal injury and neurological damage, it is important to remember that the level of cord and root injury will not coincide with that of the skeletal damage to the vertebral

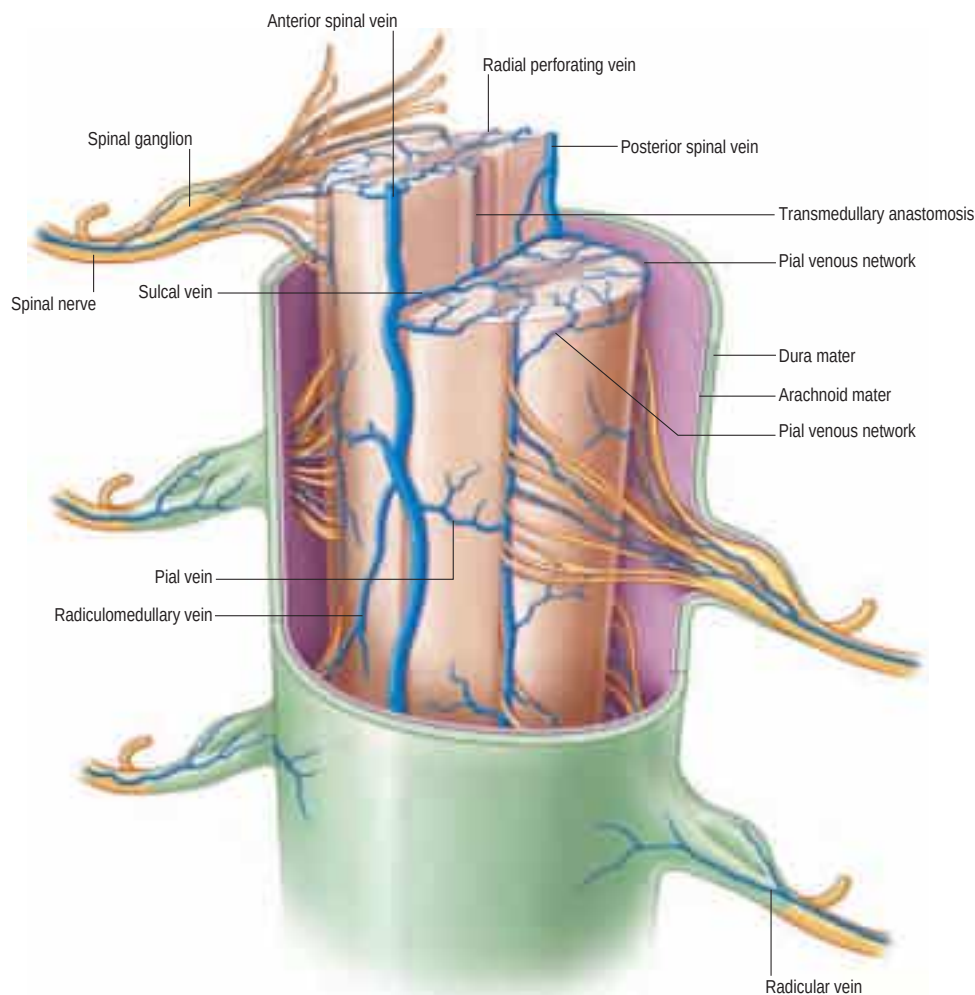


Fig. 45.14 The venous drainage of the spinal cord. (With permission from Miyasaka K, Asano T, Ushikoshi S et al 2000 Vascular anatomy of the spinal cord and classification of spinal arteriovenous malformations. *Interv Neuroradiol.* 6 Suppl 1:195–8.)

column. In estimating the vertebral levels of cord segments in the adult, a useful approximation is that, in the cervical region, the tip of a vertebral spinous process corresponds to the succeeding cord segment (i.e. the sixth cervical spine is opposite the seventh spinal segment); at upper thoracic levels, the tip of a vertebral spine corresponds to the cord two segments lower (i.e. the fourth spine is level with the sixth segment), and in the lower thoracic region, there is a difference of three segments (i.e. the tenth thoracic spine is level with the first lumbar segment). The eleventh thoracic spine overlies the third lumbar segment and the twelfth is opposite the first sacral segment. In making this estimate by palpation of the vertebral spines, it is important to remember the relationship of the individual spines to their vertebral bodies. Clinical localization may be off by several levels because other factors may come into play, such as vascular insufficiency or venous obstruction, resulting in the lesion being localized as significantly lower than it actually is (Rousseff and Tzvetanov 2006, Sonstein et al 1996).

Complete division above the fourth cervical segment causes respiratory failure because of the loss of activity in the phrenic and intercostal nerves. Lesions between C5 and T1 paralyse all four limbs (quadriplegia), the effects in the upper limbs varying with the site of injury: at the fifth cervical segment, paralysis is complete; and at the sixth, each arm is positioned in abduction and lateral rotation, with the elbow flexed and the forearm supinated, due to unopposed activity in deltoid, supraspinatus, rhomboid and the brachial flexors (all supplied by the fifth cervical spinal nerves). Upper limb paralysis is less marked in lower cervical lesions. Lesions of the first thoracic segment paralyse small muscles in the hand and damage the sympathetic outflow to the head and neck, resulting in contraction of the pupil, recession of the eyeball, narrowing of the palpebral fissure and loss of sweating in the face and neck (Horner's syndrome). However, sensation is retained in areas innervated by segments above the lesion; thus cutaneous sensation is retained in the neck and chest down to the second intercostal space because this area is innervated by the supraclavicular nerves (C3 and C4). At thoracic levels, division of the cord paralyse the trunk below the segmental level of the lesion, and both lower limbs (paraplegia). The first sacral neural segment is approximately level with the thoracolumbar vertebral junction; injury, which commonly occurs here, paralyse the urinary bladder, the rectum and muscles supplied by the sacral

segments, and cutaneous sensibility is lost in the perineum, buttocks, the back of the thighs and the legs, and soles of the feet. The roots of lumbar nerves descending to join the cauda equina may be damaged at this level, causing complete paralysis of both lower limbs. Lesions below the first lumbar vertebra may divide or damage the cauda equina, but severe nerve damage is uncommon and is usually confined to the spinal roots at the level of the trauma. Neurological symptoms may also occur as a result of interference with the spinal blood supply, particularly in the lower thoracic and upper lumbar segments.

Spinal cord injury without radiological abnormality: 'SCIWORA'



The spinal cord may be damaged without radiological evidence of skeletal injury to the vertebral column. This most commonly occurs with motor vehicle accidents but also has been reported with sports injuries and falls. MRI may or may not show evidence of cord injury after SCIWORA occurs; the majority of patients sustaining these injuries have an abnormal MRI scan.

Spinal cord syndromes



Available with the *Gray's Anatomy* e-book

LESIONS OF THE SPINAL ROOTS, NERVES AND GANGLIA

Spinal roots, nerves and ganglia may be damaged in the vertebral and root canals and at the intervertebral foramina. Neurofibromas and schwannomas may occur on the roots and nerves in the root canals. As they enlarge, these tumours become dumbbell-shaped, with both an intraspinal and an extraspinal component in continuity; the clinical picture may thus include paradoxical features as the asymmetrical space-occupying lesion grows. Root compression usually presents

Typical neural abnormalities include cord oedema, haemorrhage, contusion or transection. Typical extraneural abnormalities include intervertebral disc protrusion or herniation, ligamentum flavum bulging, cervical spondylitis, prevertebral soft tissue oedema, ligamentous abnormalities and/or ossification of the posterior longitudinal ligament. Originally, this was considered a paediatric diagnosis but it has now been shown to be equally common in adults. Spinal stenosis and pre-existing cord compression have been postulated as a risk factor for SCIWORA but clinical series have not supported this (Boese and Lechler 2013, Boese et al 2013).

For cervical spinal injury, several cord syndromes have been described, relating the clinical picture to the anatomy of the neurological lesion within the cord. The most common of these is central cord syndrome, which usually results from hyperextension injury to an osteoarthritic neck, when the major injury is to the central grey matter. This typically gives an incomplete cord injury pattern with greater motor loss in the upper extremities than in the lower limbs. There is also variable sensory loss and neurogenic bladder dysfunction (Pouw et al 2010). In anterior cord syndrome, which may occur in flexion–compression injuries of the neck, the damage occurs in the area of supply of the anterior spinal artery, sparing the posterior columns; the motor loss is usually proportionately greater in the lower than in the upper limbs, while sensory loss is less of a problem. A third syndrome that may be seen with penetrating trauma is Brown–Séquard syndrome. This is an injury directed to one side of the spinal cord, where there is ipsilateral loss of motor function and pinprick sensation and contralateral loss of pain and temperature sensation (Lee et al 1990). The anatomical basis for these symptoms is injury to the ipsilateral posterior columns and corticospinal tract, and injury to the lateral spinothalamic tract, which conveys contralateral pain and temperature sensation (see p. 306).

acutely with pain, which may be severe. The pain, paraesthesias and numbness occur in a dermatomal distribution; dermatomal overlap may make it difficult to demonstrate sensory loss on the trunk. Severe traction injuries of the upper limbs may cause avulsion of spinal roots from the cord in the cervical region.

The anatomical basis for pain of spinal origin

In the diagnosis and description of pain of spinal origin, it is particularly important to distinguish between radicular pain and somatic referred pain.

Radicular pain is caused by disorders that affect a spinal nerve or its dorsal root ganglion. This type of pain characteristically radiates, along a narrow band, into the area of the limb (or trunk wall) that the spinal nerve supplies, and is typically (although not always) lancinating in quality, not unlike an electric shock.

Referred pain is pain felt in the distribution of a nerve that is not the nerve that innervates the source of pain, although typically these nerves are derived from the same spinal cord segment. Referred pain is felt deeply across a broad area, and is usually aching in quality; although it may radiate into a limb (or around the trunk wall), it is typically constant in location. Patients may find it hard to define the outer boundaries of this pain but they can clearly identify the centre of its distribution. Referred pain can be subdivided into somatic referred pain and visceral referred pain, according to its origin. As its name implies, visceral referred pain is caused by disorders in viscera; in some cases, the referral can be quite remote from the viscus, e.g. cardiac pain that radiates into the neck. Somatic referred pain is caused by disorders of muscular and skeletal structures. Examples include lumbar spinal pain referred to the gluteal region, thigh or leg; cervical spinal pain referred to the upper limb girdle; and cervical or thoracic spinal pain referred to the chest wall, where it may mimic cardiac pain. Often the source of somatic referred pain cannot be determined by clinical examination, and invasive tests are required to pinpoint its location.

LESIONS OF THE CONUS AND CAUDA EQUINA

Lesions of the conus and cauda equina, e.g. tumours, cause bilateral deficit, often with pain in the back extending into the sacral segments and to the legs. Loss of bladder and erectile function can be early

features (Fig. 45.15). There are lower motor neurone signs in the legs with fasciculation and muscle atrophy. Sensory loss usually involves the perineal or 'saddle area', as well as other lumbar and sacral dermatomes. There may be congenital abnormalities, e.g. spina bifida (p. 759), lipomata or diastematomyelia, and the conus may extend below the lower border of L1, often with a tethered filum terminale (Fig. 45.16). Extradural lesions include prolapsed intervertebral discs. A midline (central) disc protrusion in the lumbar region may present with involvement of the sacral segments only.

Two distinct clinical syndromes, conus medullaris syndrome and cauda equina syndrome, have been described, the latter being much more common. Cauda equina syndrome typically presents in men who have a lumbar disc herniation in the setting of lumbar spinal stenosis, with symptoms of asymmetric saddle anaesthesia and asymmetric lower extremity weakness, and delayed presentation of atonic bladder and flaccid anal sphincter. Conus medullaris syndrome typically presents with symmetric saddle anaesthesia, symmetric motor deficit and earlier atonic bladder and sphincter dysfunction. The usual causes are intradural tumours or vascular lesions (Radcliff et al 2011).

Tethered cord syndrome may be associated with occult or non-occult spinal dysraphism. In its classic form, it refers to an unusually low conus medullaris (although in some 14–18% of cases the position of the conus is anatomically normal). There is some debate as to what constitutes an abnormally low conus, whether below the L1–2 disc space or below the lower border of the body of the L2 vertebra. Most commonly, tethered cord syndrome is associated with a short filum terminale more than 2 mm wide, but it also may be related to an intradural lipoma. Traction on the caudal cord and loss of filum terminale elasticity are thought to predispose to decreased blood flow in the cord, leading to symptoms (Filippidis et al 2010). The clinical presentation varies according to age. In children, the most common presentation includes foot deformities, neurological deficits, spinal deformity and various cutaneous findings such as a lumbar capillary angioma. In adults, the syndrome more commonly presents with perianal pain and leg weakness (see Fig 45.16) (Pang and Wilberger 1982).

CLINICAL PROCEDURES

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Bonus e-book images

Fig. 45.3 A T1 post-contrast sagittal lumbar spinal magnetic resonance (MR) image showing a spinal epidural abscess adherent to the lumbar dura and spinal nerves in a 28-year-old female with a history of intravenous drug abuse.

Fig. 45.4 A sagittal thoracic MR image showing a large ventral spinal epidural haematoma resulting in severe neck and upper thoracic pain and ataxia.

Fig. 45.12 Arterial disposition within the spinal cord.

Fig. 45.13 A, A sagittal thoracic MR image showing spinal cord ischaemic changes secondary to thrombotic occlusion of the artery of Adamkiewicz in a patient with a

hypercoagulable disorder. **B,** The changes were noted to be in the distribution of the anterior spinal artery, as seen on this axial image.

Fig. 45.15 A sagittal MR image showing a spinal ependymoma, a tumour arising from the filum terminale, which can be seen just below the conus medullaris.

Fig. 45.16 A sagittal MR image showing a tethered filum terminale resulting in low-lying conus.

Fig. 45.17 The lumbar interlaminar window in extension and flexion.

Fig. 45.18 A, A sagittal section of lumbar vertebrae illustrating the course of a lumbar

puncture needle through skin, subcutaneous tissue, supraspinous ligament, interspinous ligament between the spinous processes, ligamentum flavum, dura mater, into the subarachnoid space and between the nerve roots of the cauda equina. **B,** A horizontal section of lumbar vertebra illustrating the course of a lumbar puncture needle through skin, subcutaneous tissue, between the spinous processes and laminae, ligamentum flavum, epidural space, dura mater, into the subarachnoid space and between the nerve roots of the cauda equina.

Fig. 45.19 Palpation of the sacral cornua for caudal epidural injection.

Fig. 45.20 Position of the needle in caudal epidural injection.

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Fig. 45.15 A sagittal MR image showing a spinal ependymoma (arrow), a tumour arising from the filum terminale, which can be seen just below the conus medullaris. The patient presented with pain in the back radiating to the legs. At surgery to resect the ependymoma, it was found to be adherent to the filum terminale.



Fig. 45.16 A sagittal MR image showing a tethered filum terminale resulting in low-lying conus. The patient presented with back pain and perineal disturbances consistent with 'tethered cord syndrome'. The end of the conus medullaris was noted to be at the L5 level.

ACCESS TO CEREBROSPINAL FLUID

The safest approach to the cerebrospinal fluid (CSF) is to enter the lumbar cistern of the subarachnoid space in the midline, well below the level at which the spinal cord normally terminates (see above). The fine needle employed is unlikely to damage the mobile nerve roots of the cauda equina. This procedure is called lumbar puncture. It is also possible to access the CSF by midline puncture of the cerebellomedullary cistern (cisterna magna) (Ch. 18): this is cisternal puncture.

Lumbar puncture: adult

Lumbar puncture in the adult may be performed with the patient either sitting or lying on their side (lateral decubitus) on a firm, flat surface. In each position, the lumbar spine must be flexed as far as possible, in order to separate the vertebral spines maximally and expose the ligamentum flavum in the interlaminar window (Fig. 45.17). A line is then taken between the highest points of the iliac crests: this line almost always intersects the vertebral column at the L4 vertebral body or L4/L5 intervertebral disc level (see Fig. 78.16). With the spines now identified, the skin is anaesthetized and a needle is inserted between the spines of L3 and L4 (or L4 and L5). Exact identification of the level by palpation is difficult (Broadbent et al 2000). The soft tissues that the needle will traverse should also be anaesthetized; care should be taken that the injection of an excessive amount of local anaesthetic does not compromise appreciation of the structures being traversed. These will be, in order: subcutaneous tissue, supraspinous ligament, interspinous ligament, ligamentum flavum, epidural space containing the internal vertebral venous plexus, dura, arachnoid, and finally, the subarachnoid space (Fig. 45.18). The lumbar puncture needle may then be inserted in the midline or just to one side, and angled in the horizontal and sagittal planes sufficiently to pierce the ligamentum flavum in or very near the midline. There is then a slight loss of resistance as the needle enters the epidural space; careful advancement will next pierce the dura and arachnoid to release CSF. For further details about the clinical anatomy of lumbar puncture, see Boon et al (2004).

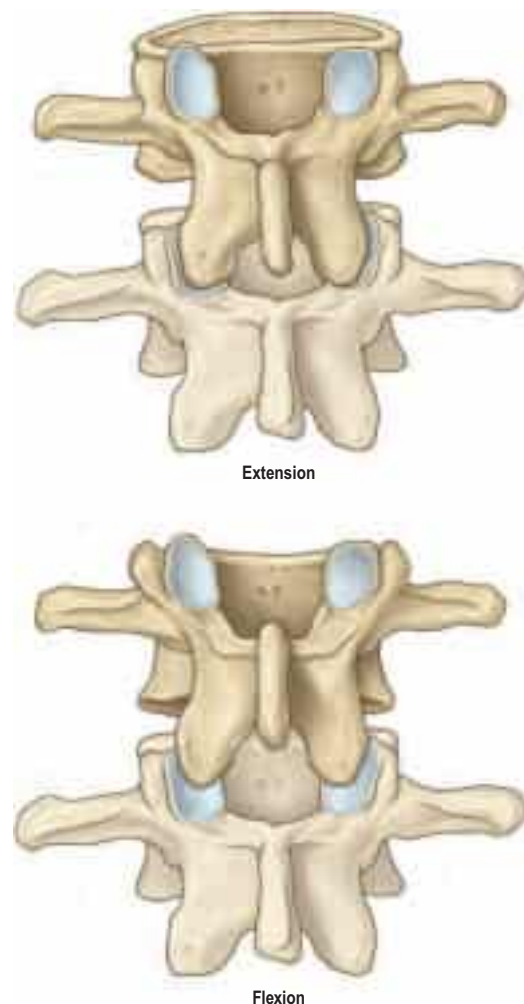


Fig. 45.17 The lumbar interlaminar window in extension and flexion.

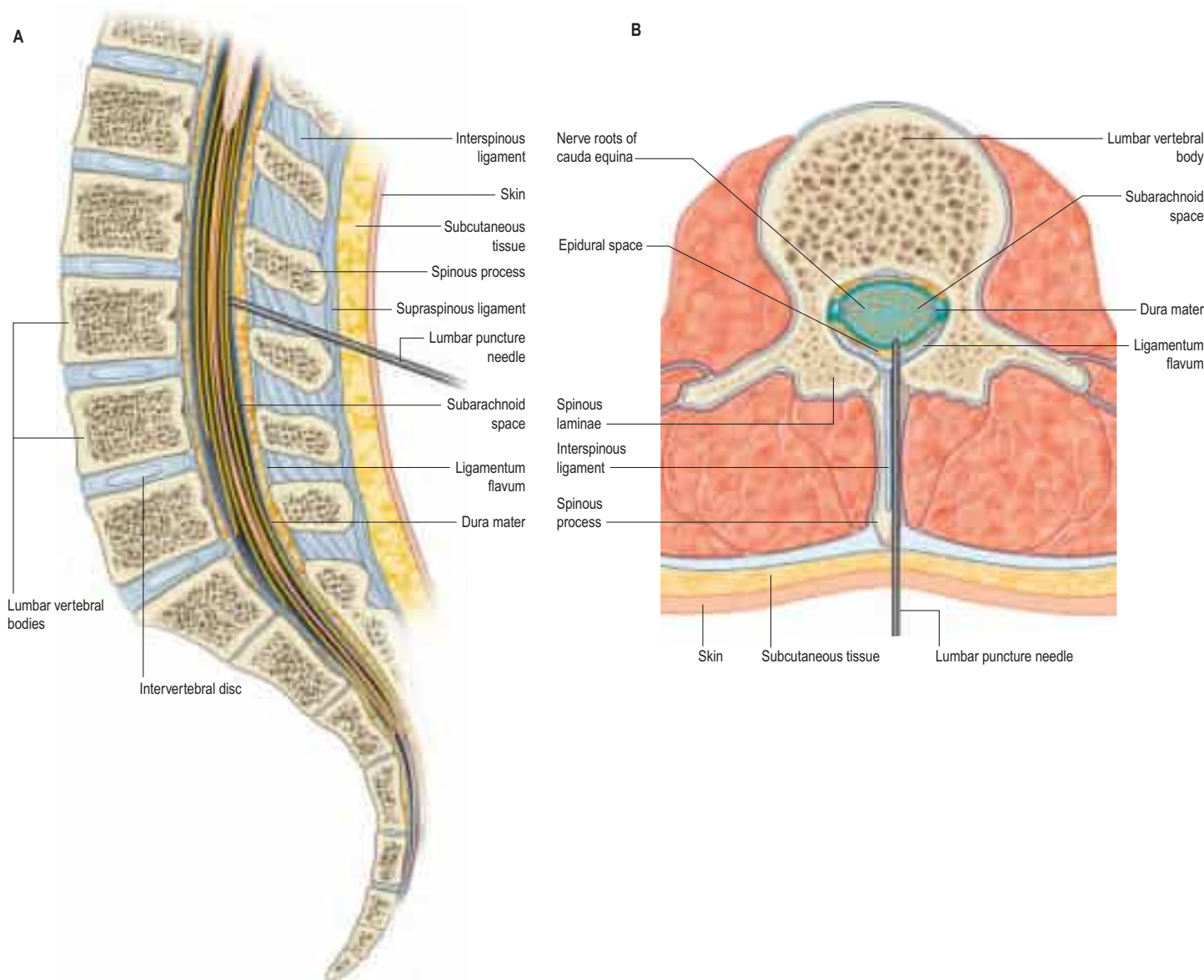


Fig. 45.18 **A**, A sagittal section of a lumbar vertebrae illustrating the course of a lumbar puncture needle through skin, subcutaneous tissue, supraspinous ligament, interspinous ligament between the spinous processes, ligamentum flavum, dura mater, into the subarachnoid space and between the nerve roots of the cauda equina. **B**, A horizontal section of a lumbar vertebra illustrating the course of a lumbar puncture needle through (in order) skin, subcutaneous tissue, between the spinous processes and laminae, ligamentum flavum, epidural space, dura mater, into the subarachnoid space and between the nerve roots of the cauda equina. (With permission from Boon JM, Abrahams PH, Meiring JH et al 2004 Lumbar puncture: anatomical review of a clinical skill. *Clin Anat.* 17:544–53.)

Lumbar puncture: neonate and infant

In premature and term neonates, the spinal cord usually terminates between the first and third lumbar vertebrae. In children between the ages of 1 and 7 years, the conus medullaris lies between the twelfth thoracic and third lumbar vertebrae (Barson and Sands 1977, Barson 1970, Vettivel 1991, Malas et al 2001, Kesler et al 2007, Suresh et al 2013). The supracristal plane intersects the vertebral column slightly higher (L3–4). These differences must be borne in mind when identifying the landmarks before undertaking lumbar puncture in the neonate and infant. A lumbar puncture is performed by placing the baby in a position, either lying or sitting, that gives maximum convex curvature to the lumbar spine. A needle with trocar is inserted into the back between the spines of the third and fourth lumbar vertebrae and into the subarachnoid space below the level of the conus medullaris. The space between L3 and L4 is approximately level with the iliac crests and it is usual to insert the needle and trocar into the intervertebral space immediately above or below the iliac crests. The distance from the skin to the subarachnoid space is approximately 1.4 cm in neonates; it increases progressively with age.

Cisternal puncture

In cisternal puncture, the cisterna magna (see Fig. 18.15) is entered by midline puncture through the posterior atlanto-occipital membrane.

Further details of this difficult specialist technique are beyond the scope of this book.

ACCESS TO THE EPIDURAL SPACE

The epidural space lies between the spinal dura and the wall of the vertebral canal. It contains epidural fat and a venous plexus. Access to this space, usually in the lumbar region, is required for the administration of anaesthetic and analgesic drugs, and for endoscopy. The caudal route is used mainly for analgesic injections.

Thoracic and cervical epidurals

It is possible to access the epidural space at thoracic and cervical levels, but the specialist techniques required are beyond the scope of this book.

Lumbar epidural

For access to the lumbar epidural space, the approach is as for lumbar puncture. The intention in epidural injection is to avoid dural puncture, so the epidural space should be entered in the midline posteriorly, where the depth of the space is greatest. The skin–epidural distance has been assessed in children at the L3–4 interspace; 1 mm/kg of body

weight is considered to be a useful guideline for children between the ages of 6 months and 10 years (Bösenberg and Gouws 1995). Techniques for entering the epidural space rely on the appreciation of loss of resistance to injection of saline (air is used by some) as the space is entered. This loss of resistance results from the fact that when the needle tip is in the ligamentum flavum, saline cannot be injected into the dense ligament, but once the epidural space is entered, injection becomes easier into the loose tissues of the space. As there is very little distance between the ligamentum flavum and the underlying dura, care must be taken not to continue advancing the needle once loss of resistance has occurred.

Caudal epidural

The route of access to the caudal epidural space is via the sacral hiatus. The space is therefore entered below the level of termination of the dural sac (the subarachnoid space and its contained CSF), which usually extends only to the level of the second sacral segment. Occasionally, the dural sac ends as high as the fifth lumbar vertebra, and very rarely it may extend to the third part of the sacrum, in which case it is occasionally possible to enter the subarachnoid space inadvertently during the course of a sacral nerve block.

With the patient in the lateral position or lying prone over a pelvic pillow, the sacral hiatus is identified by palpation of the sacral cornua (Fig. 45.19). These are felt at the upper end of the natal cleft approximately 5 cm above the tip of the coccyx. Alternatively, the sacral hiatus may be approximated by constructing an equilateral triangle based on a line joining the posterior superior iliac spines; the inferior apex of this triangle overlies the hiatus (see Fig. 78.16). After local anaesthetic infiltration, a needle is introduced at 45° to the skin, in order to penetrate the posterior sacrococcygeal ligament and enter the sacral canal. The puncture of the ligament can usually be felt as the needle is advanced. Once the canal is entered, the hub of the needle is lowered so that the needle may be fed along the canal (see Fig. 45.2; Fig. 45.20). If the needle is angled too obliquely, it will strike bone; if it is placed too superficially, it will lie outside the canal and any fluid injected can be felt subcutaneously.



Fig. 45.19 Palpation of the sacral cornua for caudal epidural injection.

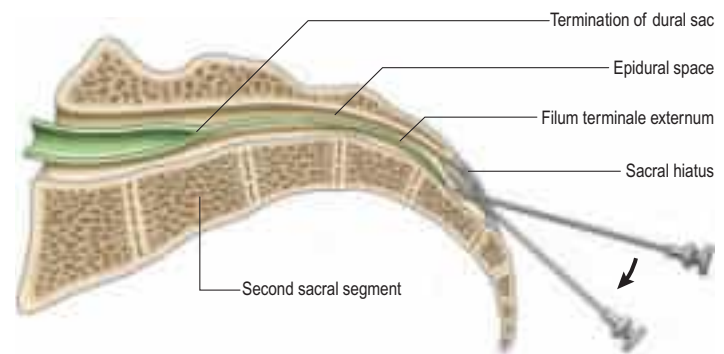


Fig. 45.20 Position of the needle in a caudal epidural injection.

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